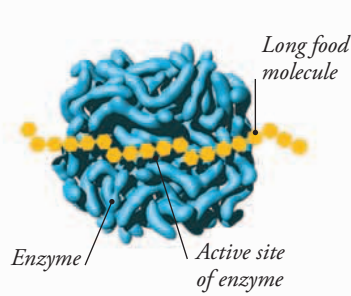
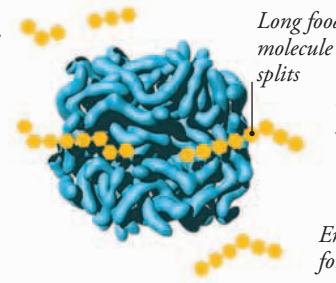


ENZYMES IN ACTION

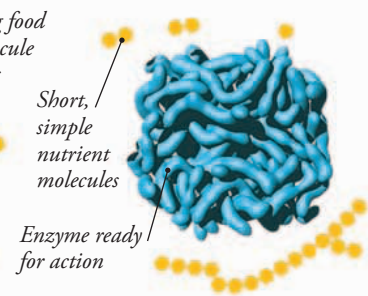
Enzymes are proteins that speed up the breakdown of complex molecules into much simpler nutrients that can pass into the bloodstream to be used by cells. Without enzymes, digestion would be so slow that we would starve. Each enzyme acts on a specific food type, such as the enzyme pepsin in stomach juice that breaks down proteins into simpler nutrients. You can see here how an enzyme works.



▲ **ATTACHMENT** Inside the digestive tract, a food molecule attaches itself to a region of its specific enzyme called the active site.



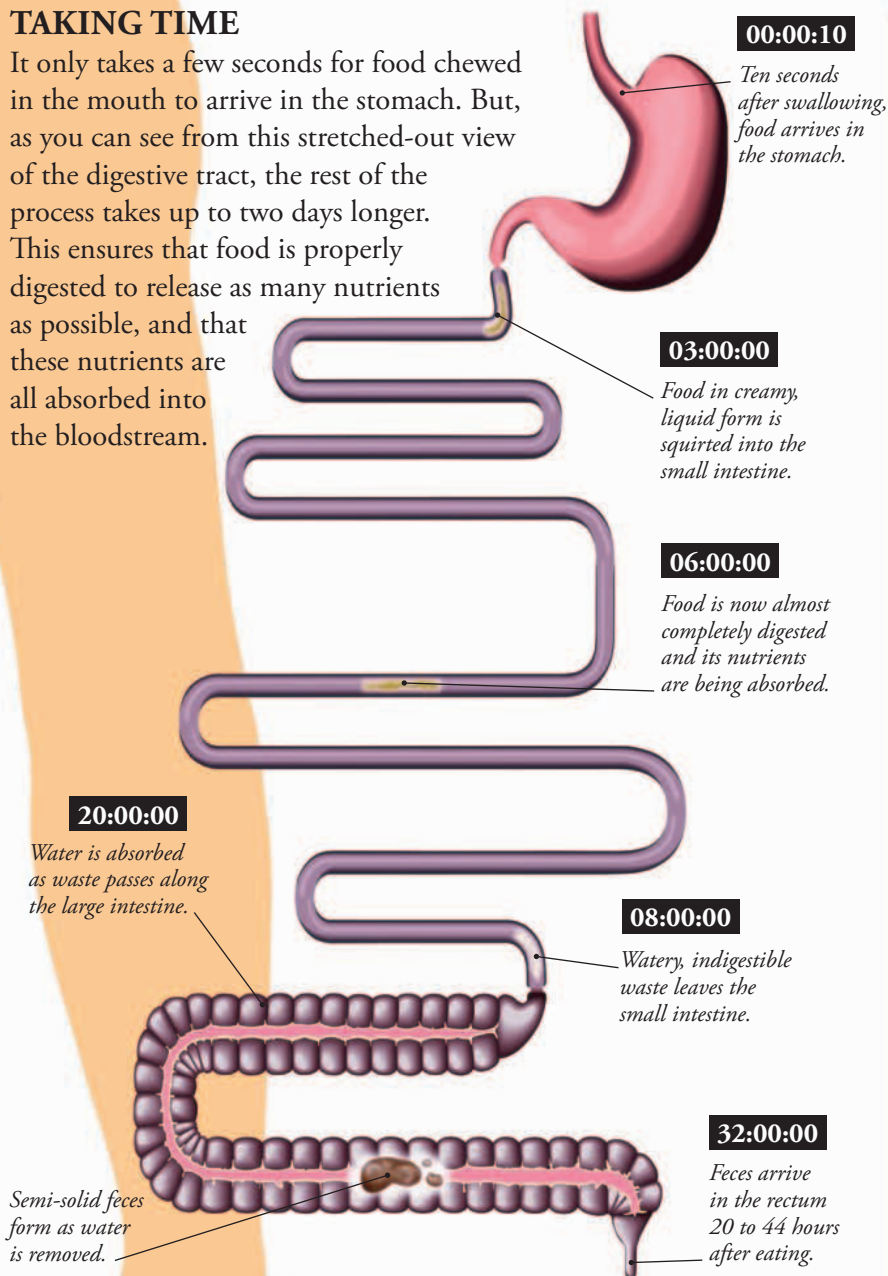
▲ **SPLITTING MOLECULES** The active site breaks some of the chemical bonds in the food molecule, splitting it into smaller nutrient molecules.



▲ **NONSTOP ACTIVITY** The enzyme releases the molecule once it has split apart. One enzyme can process hundreds of molecules a second.

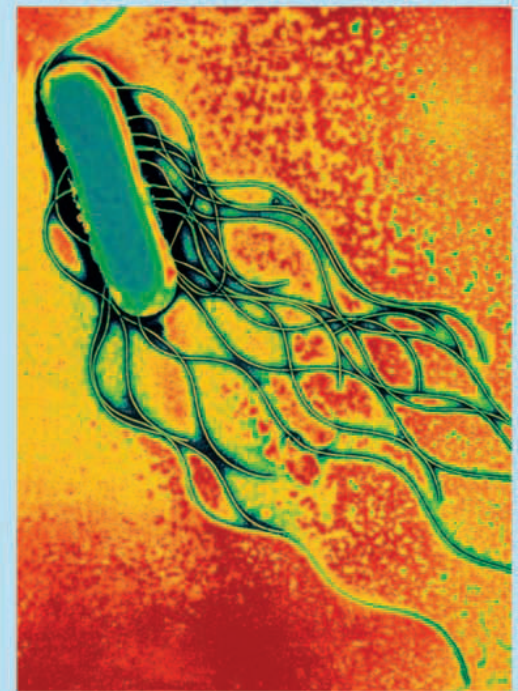
TAKING TIME

It only takes a few seconds for food chewed in the mouth to arrive in the stomach. But, as you can see from this stretched-out view of the digestive tract, the rest of the process takes up to two days longer. This ensures that food is properly digested to release as many nutrients as possible, and that these nutrients are all absorbed into the bloodstream.



LOOK CLOSER: FOOD POISONING

Most bacteria and other microorganisms in food are destroyed by the harsh, acidic conditions found in the stomach. But some harmful bacteria survive and they, or their poisons (toxins), irritate the stomach and intestines, giving you food poisoning. This causes symptoms such as vomiting (throwing up) and diarrhea (runny poo).



▲ **STOMACH BUG** This micrograph shows *Salmonella*, a bacterium that causes food poisoning. The bacterium is rod-shaped and can move by wiggling its hairlike flagella to push itself forward.

